

Final Submittal

Repair SOG OPS Facility, B3524

**Kadena Air Base
Japan**

Kadena Air Base – Tranche 1

Contract No. FA890321D0012

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CHAPTER 1 INTRODUCTION

1-1 PURPOSE

The purpose of this project is to develop a Planning Charrette Report (PCR) to meet the funding vector checks as required by current AFCEC Business Rules. This PCR will include updates to the current PCR for Kadena Air Base (KAB), meeting minutes identifying requirements, development, DD1391, and a Class III Cost estimate.

The major items within the scope of the repairs identified in this project will include replacement of the facility roof and mechanical and electrical upgrades.

1-2 AUTHORIZATION

1-2.1 Scope of Work

This project will repair the facility's structural concrete spalling, roof, and mechanical and electrical systems. The concrete roof will be completely rebuilt, including an anchoring system and seismic improvements. Insulation will be installed on the interior side of the exterior walls and below the roof slab. The electrical, ventilation, air conditioning, and fire protection systems will be replaced throughout the building. The central communication room will be relocated to another facility. The project also includes the patching and repair of concrete spalling, structural repairs to the concrete masonry unit load-bearing walls, and repairs to finishes including gypsum and drop ceilings.

Temporary facilities are not needed during execution.

1-3 APPLICABLE CRITERIA

The most current references listed below at the time of design will apply.

DAFI 32-1020	Planning and Programming Built Infrastructure Projects, with Change 1, 19 January 2022
FY25-29 AFCAMP	Business Rules
UFC 1-200-01	DoD Building Code, with Change 3, 26 February 2024
UFC 1-200-02	High Performance and Sustainable Building Requirements, with Change 2, 01 June 2022
UFC 1-300-02	Unified Facilities Guide Specifications (UFGS), with Change 3, 18 May 2021
UFC 3-101-01	Architecture, with Change 4, 08 January 2024
UFC 3-110-03	Roofing, with Change 5, 12 June 2024
UFC 3-120-01	Sign Standards, with Change 4, 08 January 2024
UFC 3-201-01	Civil Engineering, with Change 6, 20 December 2022
UFC 3-210-10	Low Impact Development, with Change 3, 01 June 2015
UFC 3-220-01	Soils Investigation Sampling, 1 November 2012
UFC 3-301-01	Structural Engineering, with Change 1, 02 October 2023

UFC 3-401-01	Mechanical Engineering, with Change 1, 01 October 2015
UFC 3-410-01	Heating, Ventilating, and Air Conditioning Systems, With Change 9, 09 January 2024
UFC 3-420-01	Plumbing Systems, 01 April 2021
UFC 3-501-01	Electrical Engineering, with Change 2, 15 December 2023
UFC 3-520-01	Interior Electrical Systems, with Change 2, 12 April 2021
UFC 3-530-01	Interior and Exterior Lighting Systems, with Change 5, 09 February 2023
UFC 3-560-01	Operation and Maintenance: Electrical Safety, with Change 1, 24 July 2017
UFC 3-580-01	Telecommunications Interior Infrastructure Planning and Design, With Change 1
UFC 3-600-01	Fire Protection Engineering for Facility, with Change 6, 06 May 2021
UFC 4-010-01	DoD Minimum Antiterrorism Standards for Buildings with Change 3, 24 May 2024
UFC 4-021-01	Design O&M: Mass Notification Systems, with Change 1, 01 January 2010
ABA	Architectural Barriers Act
ASCE 7-16	Minimum Design Loads and Associated Criteria for Buildings and Other Structures
ASCE 41-17	Seismic Evaluation and Retrofit of Existing Buildings
ASHRAE	Energy Standard, 2019 Edition
ECB 2018-17	New Requirements for Visual Notification for Mass Notification Systems, with Revision 3, 31 August 2023
EM 1110-2-2502	Engineering and Design Retaining and Flood Walls, 29 September 1989
HDS-5	Hydraulic Design of Highway Culverts Third Edition, April 2012
IBC	International Building Code, 2021
IEBC	International Existing Building Code, 2021
IMC	International Mechanical Code, 2021
IPC	International Plumbing Code, 2021
Kadena Air Base	Design Guide, 07 February 2020
NFPA 1	Fire Code, 2024 Edition
NFPA 13	Standard for the Installation of Sprinkler Systems, 2022 Edition
NFPA 24	Standard for the Installation of Private Fire Service Mains and Their Appurtenances, 2022 Edition
NFPA 30	Flammable and Combustible Liquids Code, 2024 Edition
NFPA 70	National Electrical Code, 2023 Edition
NFPA 70E	Standard for Electrical Safety in the Workplace, 2024 Edition
NFPA 72	National Fire Alarm and Signaling Code, 2022 Edition

NFPA 101	Life Safety Code, 2024 Edition
NFPA 110	Standard for Emergency and Standby Power Systems, 2022 Edition
RP 10	Standards of Seismic Safety for Existing Federally Owned and Leased Buildings
TR-55	Urban Hydrology for Small Watersheds, June 1986

1-4 PROJECT DESCRIPTION

[REDACTED]

1-4.1 COA 1: Roof Replacement

COA 1 would focus solely on the roof replacement for Building 3524 (B3524). Minimal changes to the walls and foundations will occur to support the roof replacement and move HVAC equipment off of the structure, but the design will be limited to 25% of structure replacement imposed by DAFI 32-1020 section 3.3.3.3 on FSRM Repair-by-Replacement. The funding source for COA 1 would be FSRM.

The Maximum Construction Costs (MCC) are not anticipated to be greater than 50% PRV and will thus not trigger AT/FP requirements and code updates. Additional engineering work will be completed on items that are only necessary with the replacement of the roof.

MCC is anticipated to be greater than 30% PRV, so the project does trigger ASCE 41 seismic analysis per RP 10. In addition, an analysis of roof framing for high winds is triggered per IEBC. This analysis should be performed for the new roof and overhangs.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

1-5 PROJECT SITE

B3524 is a single-story, 38,875 square feet (3,612 square meters) concrete building located on the eastern end of KAB. The building was constructed in 1967 as an administrative facility and consists of open and closed administrative workspace; an auditorium; an operations center; storage; a break room; and mechanical, electrical, and telecommunications rooms. In 2023, extensive concrete spalling from the ceiling caused major safety concerns and led to the evacuation of the facility until repairs could be made, although some critical equipment functions remain such as a telecommunications room that serves multiple facilities.

Surveys were completed in 2022 and 2024 to analyze the extent of structural damage to the concrete and reinforcing bars of the facility as well as the properties of the concrete to determine the best repair methods. The concrete has become carbonated due to the age of the structure, poor building envelope, prolonged exposure to the elements, and inadequate preventative maintenance. The process of carbonation is irreversible and difficult to stop in the presence of moisture, which means that the spalling will continue even if surface repairs such as patching are made. This also makes the facility more vulnerable to failure of the structural system in a major event such as a typhoon or earthquake.

In addition to the spalling issues, there is significant cracking in the concrete masonry unit (CMU) walls at the top of the three rooftop penthouses, significant leaking around the central workspace, and the roofing material is worn and beyond its service life which has led to the described water infiltration causing other structural damage.

In addition to the concrete repairs, the electrical and mechanical systems in the building require replacement.

The structure is currently mostly uninhabited due to the safety concerns posed by the concrete spalling however, some of the critical equipment functions that are housed in B3524 still remain.

There was an asbestos containing material (ACM) and lead based paint (LBP) survey that was completed of portions of the building in 2020 that did not find any ACM or LBP. However, given the age of the building and prevalence of ACM and LBP in local construction, there is a possibility that other areas of the building do have these materials present.

Figure 1-1 Kadena Air Base Project Location (Image Source: Google Earth)



Figure 1-2 SOW OPS Facility, B3524 (Image Source: Google Earth)



1-5.2 Functional Objective

If this project is not provided, [REDACTED]

[REDACTED] This would restrain space in an already limited installation facility space for an ever-growing workforce at KAB, especially when many other facilities require spalling repairs as well.

1-5.3 Post-Construction Considerations

Contractors must perform thorough field investigations of the facility both before and after the construction phase to ensure compliance with government requirements. The contractor will be held accountable for any damage occurring during construction and must address any damage identified. Additionally, the contractor is required to submit a detailed work schedule for the building during the performance period. R & O will be present for the final inspection to confirm that all government requirements have been satisfied.

1-5.4 Personnel and Equipment

This facility directly supports the 353 SOW mission to support special operations aviation activities throughout the Pacific and protect U.S. interest in theater.

[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]
[REDACTED]	[REDACTED]
	[REDACTED]



CHAPTER 2 CIVIL

2-1 GENERAL

Per observed drainage patterns of the existing topography and tributary area delineation, significant offsite runoff, and runoff from the east side slope of the building is collected in the existing swale running along the north side of B3524. The 5 feet width x 3.5 feet height earthen berm with masonry wall adjacent to the swale functions as a floodwall diverting the offsite runoff downstream. There are three (3) openings along the earthen berm for access to the building. The finish grade of the opening is flush with the adjacent grade of the slope, enabling off-site runoff to enter the site. Installing barrier berms at the openings would be a cost-effective flooding control measure to eliminate additional off-site runoff entering the site and accumulating in the existing swale.

Observed watermarks appear at the open area between the edge of the building and the earthen berm north of the building. Temporary ponding of stormwater occurs in the area due to limited capacity of the existing earthen swales and drainage system.

In addition to stormwater detention at existing swale, the adjacent B3531 located downstream has flooding issues during major storm events. The downstream storm drains discharge to a nearby creek next to Halyburton Street, and watermarks are observed at the catch basins at the parking lot upstream of the discharge point.

Most likely, the downstream storm drains are undersized to handle the peak flows, and the surcharged storm drain reached its capacity causes stormwater detention within the local depressed area between the building and the sloping berm.

2-2 APPLICABLE CODES

See Section 1-3.

2-3 BASIC DESIGN INFORMATION

Hydrology and hydraulic analysis shall be performed to estimate the peak flows that contribute to the site for storm drain system upgrade and flood control.

Evaluate and determine the Best Management Practice (BMPs) system with onsite stormwater detention system, flow diversion, or increase storm drain system capacity downstream.

2-4 CALCULATIONS

Topography survey data for drainage area delineation (both off site/on-site).

Perform hydrology analysis for peak flows estimation of targeted design storm events per TR-55.

Perform hydraulic analysis to determine/verify capacity of existing storm drain system per HDS-5.

CHAPTER 3 ARCHITECTURAL

3-1 GENERAL

B3524 is a facility that houses multiple headquarters functions, [REDACTED], a joint operations center, and an assembly area. A walkthrough of the facility was completed to primarily address the concern of concrete spalling in the interior and exterior of the building. It was confirmed that the facility is experiencing extensive concrete spalling in the roof slab due to corrosion expansion of reinforcement. Currently the facility is mostly vacated. A few occupants and functions remain active in some areas and there is a communications room that serves other facilities.

3-1.1 Applicable Codes

See Section 1-3.

3-2 BASIC DESIGN INFORMATION

Three COAs have been identified and include:

- **COA 1:** Demolition of the existing roof slab and roof beams, leaving all other load bearing walls and structure in place or modified as needed for floor plan efficiency.

[REDACTED]

[REDACTED]

3-2.2 COA 1: Roof Replacement

This COA will demolish the roof structure and roof beams and all other structural members to remain in place.

This is considered a partial reconstruction of the existing facility. The overall layout regarding circulation and main floor areas will remain similar to the existing layout, since the facility will have to match the existing shear walls, load bearing walls and column locations. The existing layout cannot be modified much and will have to match the main blocks, one of them being the auditorium, which may have to be assigned to other functional areas. [REDACTED]

[REDACTED]. All mechanical and electrical rooms must be on the ground floor. The structure will be reinforced concrete throughout, including the roof deck. Insulation will be installed on the interior side of the exterior walls and below the roof slab. Interior walls will be primarily of gypsum wall board construction [REDACTED]

[REDACTED] The telecommunications node room will be protected and not relocated to another facility.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Refer to the APPENDIX A for drawings SD101 for COA 1, [REDACTED]

[REDACTED]

CHAPTER 4 STRUCTURAL

4-1 GENERAL

4-1.1 Existing Conditions

The existing structure was constructed around the year 1967. It is a one-story building with a low sloped reinforced concrete roof supported by concrete beams and columns. The perimeter walls and most interior walls are CMU load bearing shear walls. The walls bear on shallow concrete strip footings. Shallow spread footings support concrete columns and are connected to adjacent wall strip footings with tie beams. There is a large ground floor mechanical room with exterior access, and three mechanical rooms in CMU-walled penthouses on the roof. The portion of roof above the auditorium is constructed of long span concrete beams, upturned so the tops of the beams extend up above the roof slab. The center section of the building is similarly covered by upturned concrete beams. In addition, that roof slab was constructed to be lower than the surrounding roof, with sloping surface and drainage holes through the upturned beams all leading to one center drain. A corrugated galvanized steel roof was added in 1982 above that roof to shed water.

The existing structure has extensive concrete spalling in the roof slab due to corrosion expansion of reinforcement. The spalling is significant in the un-conditioned mechanical room, and in Rooms 188 and 120-121. There are smaller spalls and concrete delamination at several other locations. Fiberboard insulation is fastened to the underside of the roof slab, holding moisture to the roof, and hiding any spalling. Most interior spall locations were observed located below small roof penetrations, areas with ponding, and/or rooftop equipment anchor points. The spalling appears to be limited to the roof slab, with minimal spalling observed in beams and columns. The exception to this is the concrete beams in non-conditioned spaces, such as the mechanical room and penthouses were observed to have significant deterioration and spalling. The CMU shear walls appear to be in structurally acceptable condition. The slab-on-grade appears to be in satisfactory condition with one exception where the slab appears to be cracked and bulging up.

The spalling issue has caused the occupants of the building to be evacuated to other locations and the building is mostly vacant. Samples of the roof slab concrete at this building were obtained in January 2024 and have undergone laboratory testing to determine the concrete properties. Testing of the samples includes chloride content, petrography, compressive strength, and carbonation. A testing report has been produced for this investigation which states the structural deterioration is mainly attributed to carbonation of the concrete. The concrete has become carbonated due to the age of the structure, a poor building envelope, prolonged exposure to the elements and a lack of preventative maintenance. The process of carbonation is irreversible and difficult to stop in the presence of moisture. For perspective, it could take approximately two years for the roof slab to dry out if the roof envelope is repaired and the underside of the roof slab concrete is exposed to conditioned air. As such, additional spalling and deterioration is anticipated.

These spalls can injure occupants and equipment below. In addition, this loss of concrete thickness and reinforcement cross section results in a loss of flexural and shear capacity of the concrete slab. Over time such loss will progress to cracking, allowing more water penetration and deterioration, leading to additional concrete section loss. This increases the chances of a possible failure of the structural system during an event such as a heavily loaded maintenance repair, typhoon, or earthquake.

In addition to the concrete spalling issues, there is significant cracking in the CMU at the three rooftop penthouses. There is also significant leaking where the lowered center section roof meets the walls at the surrounding hallway.

The roofing is a reinforced, fluid-applied membrane. The roofing membrane exhibited signs of wear, brittleness, and has reached the end of its useful life.

4-1.2 New Work

Structural work will vary based on the COA that is chosen. See section 4-2 below. Scope will involve designing and detailing new concrete, steel, and CMU construction, and repairs to any existing to remain. Nonstructural systems including Ceilings, Piping, and Mechanical systems will require seismic bracing.

4-1.3 Applicable Codes

See Section 1-3.

4-2 BASIC DESIGN INFORMATION

Due to the spalling issues, New Work can involve:

4-2.1 COA 1: Roof Replacement

Demolish the entire roof and concrete beams down to top of existing CMU walls - leaving walls, slab on grade, and footings in place. Where wall heights vary, new CMU or concrete can be added to achieve a consistent elevation. The roof will be rebuilt while keeping existing concrete columns and CMU shear walls in original locations above existing grade beams and footings.

The roof can be replaced with a new cast-in-place concrete slab; precast structural elements, such as hollow core slab or double tees topped with a concrete topping; or steel joists and metal deck. New connections and rebar will be doweled into the existing CMU and concrete, using an adhesive anchoring system.

The design and construction of the existing CMU walls, concrete columns, and footings will be evaluated to ensure that they are not overloaded and can meet the lateral load capacity requirements of IEBC, AISC 41, and RP 10. New shear walls, columns and footings may need to be added as required by the Code. Select interior walls can be demolished as needed by Architectural requirements, but for the most part the existing wall layout will remain. The existing mechanical penthouses will be demolished and

Concrete beams and roof slab will be designed and constructed at a consistent plane that slopes to the roof edges, eliminating the existing central lowered slab and raised metal roof. The consistent elevation will allow for water to drain to edges unimpeded. Over the central area and auditorium, concrete beams will be designed to be shallower and with bottom of beam extending below slab. Or a metal roof deck diaphragm can be designed to be supported by structural steel girders and beams.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



4-2.4 Load Data

- Risk Category: IV per UFC 3-301-01, Table 2-2 for “Building having DoD mission-essential command, control, primary communications, data handling, and intelligence functions as designated by the using agency”
- Wind: 190 mph (306 kph) Exposure C
- Tsunami: Not in Tsunami Inundation Zone
- Ground Snow Load Pg: 0 psf (0 kPa)
- Seismic: $S_s = 1.73g$, $S_1 = 0.93g$
- Peak Ground Acceleration = 0.66g
- Site Class D (default assumed pending geotechnical engineering investigations)
- Seismic Design Category: F

CHAPTER 5 MECHANICAL

5-1 GENERAL

The SOW Operations Building, B3524, is experiencing major structural degradation. There are three (3) COAs that are being evaluated for addressing the structural issues and all three (3) COAs involve the complete replacement of the HVAC and Plumbing systems in the existing facility or providing all new HVAC and Plumbing systems in a new facility. Based on the current condition of key HVAC and plumbing components, the replacement of much of these systems is indicated, regardless of the structural situation. Although some components have been replaced recently, those components will likely be repurposed at other facilities based on the timeline for reconstructing or replacing this building. Currently, there are four air handlers on the roof in penthouse structures. There is a desire to move all air handling units to the ground, for ease of maintenance and protection during storms. This will require the construction of ground level Mechanical Rooms, either within the existing building footprint or as additions to the building. There is no available space on the ground on the south and west sides of the existing building footprint for new additions, and there is limited space on the east side of the building. Most of the new Mechanical Room construction is limited to the north side of the building or new Mechanical Room space can be set aside within the existing building footprint.

5-2 FUNCTIONAL AND TECHNICAL REQUIREMENTS

5-2.1 Applicable Codes

See Section 1-3.

5-2.2 Design Temperatures

HVAC loads should be based upon the local design weather values. For Kadena Air Base on Okinawa, Japan the following values are recommended:

Outdoor Design:

Summer Conditions:	91.3°F dry bulb	ASHRAE 1% DB
	80.3°F wet bulb	ASHRAE 1% MCWB
Winter Conditions:	50.3°F dry bulb	ASHRAE 99.6% DB

Indoor Design:

Summer:	78°F dry bulb
	50% relative humidity
Winter:	68°F dry bulb
	0-50% relative humidity

5-2.3 Environmental Considerations

Kadena Air Base is in an Environmental Severity Classification (ESC) zone of C-5, the most severe/highly corrosive environment. Any new design should include a Corrosion Prevention and Control (CPC) Plan that details implementation of the mandatory requirements of UFC 1-200-01, Chapter 4 & Appendix A. The CPC Checklist Tool from the Whole Building Design Guide (wbdg.org) website should be utilized to ensure compliance.

A new survey for Lead Based Paint (LBP) and Asbestos Containing Materials (ACM) shall be performed, while referencing the existing LBP/ACM report from August 2020.

5-3 DESIGN OBJECTIVES AND PROVISIONS

The existing building is conditioned by two 60-ton air cooled chillers that were replaced in 2023. There is a series of three chilled water pumps, with two being newer than the third. Chilled water is pumped to one air handler in the ground floor Mechanical Room and up to the roof to four air handlers in three Penthouses. These five air handlers serve the four quadrants of the building and center area, including [REDACTED] non-secure office areas. Four air handlers are dated 2005 and one is dated 2011. The restrooms are served by local exhaust fans ducted to the outside. All COAs being considered will result in the complete demolition of the existing HVAC system, except for the chillers.

Any new HVAC system will need to follow the requirements of UFC 1-200-02 and have a life cycle cost estimate performed comparing the first cost, maintenance costs and efficiency of multiple system types to determine the most advantageous energy using systems for this building. Any new HVAC system will likely utilize chilled water with multiple air handlers in several Mechanical Rooms, separate Dedicated Outdoor Air Systems (DOAS) to precondition outside air, Variable Air Volume (VAV) type air distribution and Direct Digital Controls (DDC). The chilled water system shall try to reuse the existing chillers while also providing a fully redundant chiller/pump system. As part of COA-1 & COA-2, the existing Information Transfer Node (ITN) shall remain during construction and be provided with all necessary utilities, including HVAC. All exterior HVAC equipment should be sheltered in screen walls with a chain link fence cover for protection from flying debris during severe weather events. [REDACTED]

[REDACTED] New exhaust systems will be provided from restrooms, and other areas, and be ducted out the exterior walls. Ductless split systems, or Computer Room Air Conditioners (CRAC), will be provided for any Telecomm Rooms, depending on the size. In addition, a VAV box from the central HVAC system should be provided to all Telecomm Rooms for redundancy. New DDC controls will be provided and can be tied into the base wide BACnet control system.

The existing plumbing system has a combined fire and domestic water service entrance backflow located on the southwest corner of the building that comes into a storage room at that location. There is a second domestic water only service that comes into the building through the north side mechanical room. The current sanitary sewer exits the building through four connections, one on the north side, two on the east side and one on the south

side. There are existing issues with sewer flow within the building, with some underfloor sanitary sewer lines suspected of having sunk into the ground causing back-ups. There are currently Men's and Women's group toilets in the building's southwest area, but only a single Men's room in the northeast area. Based on system condition, age and the proposed COAs extensive construction impacts, all existing plumbing within the building footprint will be removed, including all under floor piping.

Any redesign should address the inequity of restrooms available for all occupants. Domestic and sanitary sewer connection points will be provided from the utilities surrounding the building. All new domestic cold and hot water piping will be provided below the floor, in walls and above the ceilings. All new sanitary sewer piping will be provided below the floor, in walls and above the ceilings, with venting through the roof. All restroom fixtures will be new in any restoration and modernization of the existing building, including water closets, urinals, lavatories, and showers. New water fountains will be provided at restroom locations. New electric tank type or instantaneous water heaters will be provided where hot water is required.

Seismic restraints will be required for equipment, piping, and ductwork in the building. The Seismic Design Category, Risk Category and Site Class as determined by the Structural Engineer must all be considered when determining which components need seismic restraints.

Provide complete O&M information at the end of the project, including warranty information for all new equipment.

5-4 CALCULATIONS

A detailed analysis of the building is required to determine the final sizing of the new HVAC equipment. This analysis will use local weather data and information about the number of people, equipment, and lighting in each room. The building envelope construction will be updated as part of the building reconstruction and used in the HVAC load calculations. The detailed analysis will determine the equipment size required, and the appropriate distribution of the cooling to all areas of the building. Loads will be calculated using a computerized program that uses ASHRAE Fundamentals methods of calculation. High Performance and Sustainable Building (HPSB) practices, as outlined in UFC-1-200-02, must be followed.

5-5 COORDINATION WITH INSTALLATION OR OUTSIDE AGENCIES

The building will have a DDC system that communicates using the BACnet protocol and will communicate to the base-wide Energy Monitoring and Control System (EMCS) over the Base CE VLAN to Building 1474 (B1474). Cybersecurity will be integrated into the DDC system, especially for the tie-in to the EMCS. Utility meter monitoring will be included per the requirements of the Kadena Design Guidelines.

CHAPTER 6 ELECTRICAL

6-1 GENERAL

6-1.1 Existing Conditions

The facility's existing electrical system consists of a 120/208V, 3 Phase, 4 Wire, 2000A switchboard powered from a 500 kVA unit substation with a 1200A circuit breaker located just outside of the building.

The switchboard consists of Bus A, for normal building loads, and Bus B, for emergency building loads. Bus A is powered from the 500 kVA unit substation via a 1600A main circuit breaker in the switchboard. Bus A is connected to a 208V, 600A, 4 Pole, 60 Hz, automatic transfer switch (ATS) via a 600A circuit breaker.

This ATS is connected to a 150 kW, 120/208V, 3 Phase, 4 Wire, 60 Hz, diesel emergency standby generator and to the Bus B side of the switchboard via lugs only.

Switchboard Bus A provides power to a wire gutter, which provides power to rooms 116, 135, 178, and 193, and to motor control center (MCC) MCC A, which powers chiller units, chilled water pumps, and an air handling unit.

Switchboard Bus B provides power to several panelboards and air handling units, and provides power to MCC B, which provides power to the fire alarm system, roof vent, and a few existing loads. Bus B is the facility's emergency circuit, providing power to all areas of the SCIF and parts of the JAOC area and HVAC system. Unfortunately, not all areas of the JAOC and HVAC system are powered from the emergency circuit, as would be preferred.

The switchboard, MCC A, and MCC B are in extremely poor condition and pose a serious safety hazard for any personnel working on or near the equipment. The equipment is covered in rust with large holes in the front plates. Exposed electrical wires, circuit breakers, and more can be seen and easily reached. Some electrical compartment doors on the motor control center are unable to close.

It is unclear exactly how old the switchboard and MCCs are, but they are over 30 years old and may be around 50 years old. Modifications in 2011 replaced the old generator with the current 150 kW Caterpillar generator and 600A Caterpillar ATS. Based on visual inspection and age, the ATS and generator appear to be in fair condition and could be reused or repurposed. The 1,000-gallon diesel fuel tank, located outside, was not replaced during the modification project.

Panelboards throughout the facility appear in acceptable condition due to being in a climate-controlled environment but are likely close to the end of their useful life.

While light controls appear older, during the site visit, it appears all fluorescent lights were replaced with LED lights.

No arc flash labels are present on any electrical equipment, which is in violation of UFC 3-560-01. This means, it is not clear what level of protective personnel equipment (PPE) is required to work on the electrical equipment, if energized work/maintenance is required (highly likely based on the importance of some electrical equipment), and this can lead to serious injuries or death from an arc flash event.

6-1.2 New Work

COA 1 will require the replacement of the roof mounted electrical conduits and conductors and temporary removal of lights. The lights appear to be LED and recently replaced, so they can be reinstalled after the new ceiling is installed. The conduits and conductors should be replaced as they are too old to warrant removal and reinstallation.

Due to the existing condition of the switchboard and motor control centers, there is no benefit to reusing or repurposing this electrical equipment when rewiring with the new conductors. This equipment will be replaced with a new 120/208V, 1,200A, main switchboard “MP”, and a 120/208V, 600A, emergency panel “EP” for emergency loads. Additional new electrical equipment is required to provide power to the new HVAC units being installed to replace the old HVAC units.

The existing 150 kW generator, 600A ATS, and 1,000-gallon fuel tank can be reused for the new emergency panel “EP”.

A power system study, including an arc flash analysis, must be conducted for the new electrical distribution equipment, including any switchgear, switchboards, motor control centers, and panelboards installed in the building. The arc flash study must include the production and installation of arc flash labels in compliance with UFC 3-560-01.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

6-1.3 Applicable Codes

See Section 1-3.

6-2 CALCULATIONS

The calculations for electrical design elements, such as those listed below, will utilize metric units. If the project is permitted to use IP units, the calculations shall be performed in normally accepted and recognized IP units.

- Maintained lux [Foot candle (FC)] levels in all areas.
- Individual circuit and system loads tabulated in amperes for each panelboard or switchboard.
- Generator sizing calculations to verify the existing generator is appropriately sized or to size a new generator.
- Transformers, panelboards, switchboards, and feeders indicating all demand, diversity, and ambient-temperature or conductor-grouping factors considered in the selection of equipment or conductor sizes.
- Voltage-drop calculations as necessary for long distance connections.
- Weight, dimensions, and electrical characteristics of each major item of equipment supported by manufacturer's names, and catalog and model numbers.
- Short circuit calculations.

6-3 COORDINATION WITH INSTALLATION OR OUTSIDE AGENCIES

6-3.1 Telecommunications Node

The building contains a telecommunications room, ITN, which acts as 1 of 13 communication nodes on the base. Many nearby facilities are connected to and rely on this telecommunications node.

The ITN telecommunications node poses a serious problem for any major modifications of this structure. For COA 1 and 2, if the building is completely retrofitted, requiring stripping the building down to its foundation and bare bones, the ITN telecommunications node would need to be protected and powered the entire time during construction.



All construction that would affect this node must be coordinated with the base's telecommunications team.

6-3.2 Other Coordination Considerations

Coordination with the electrical utility is required to turn off the upstream unit substation circuit breaker prior to construction, demolish from and reconnect to the new system, and reenergize the system once construction is complete.

CHAPTER 7 FIRE PROTECTION AND LIFE SAFETY

7-1 GENERAL

B3524, built between 1967-1968, is a single story reinforced concrete structure with 8-inch-thick, reinforced CMU walls. [REDACTED]. A site walk was performed to primarily address the concern of concrete spalling in the interior and exterior of the building. It was verified that the facility is experiencing extensive concrete spalling in the roof slab due to corrosion and expansion of structure reinforcement. Currently the facility is mostly vacated.

7-2 EXISTING CONDITIONS

The type of construction in the existing building is non-combustible type IB per IBC definitions, with interior metal stud walls with non-combustible concrete/cement coverings. The types of occupancy classifications in the building include Business (B), Factory moderate hazard (F), and Storage (S), per IBC and Life Safety Code definitions. The occupancy classifications in the building are classified as low, and ordinary hazards (L, OH). The existing fire suppression system consists of a wet sprinkler system serving all areas of the building. The building is considered a fully sprinkler building. The building also utilizes a conventional fire alarm system. Add fire alarm

7-2.1 Fire Flow Test Results

A fire flow test per NFPA and UFC 3-600-01 guidelines was conducted on January 24, 2024, to verify the strength of water supply source. The results from the fire hydrant test were used to analyze the available fire flow per Fire Code requirements and are included in this report as APPENDIX D. The test results indicated 92 psi static and 44 psi residual pressure at 856 gpm flow. This equals to a fire flow of 1,066 gpm at 20 psi and suggests an average towards low water supply source character for expected fire hazards.

7-2.2 New Work

To address the spalling and structural issues, three COAs have been proposed and are listed in Section 1-4.

In all three proposed COAs, the fire suppression systems will be demolished entirely and replaced with new ones per current code requirements. [REDACTED]

[REDACTED] However, per COA1, the wall mounted fire detection, alarm, and control equipment will remain intact, but conductors and protecting raceways and cable trays will be completely removed and replaced. This is due to the fact that all conductors, raceways, and cable trays are supported by building structural elements (roof), and with removal of the roof, all supported raceways and cable trays will be demolished as well.

The new building layout will be equivalent to the existing facility for appropriate configurations and performing similar functions. The new building consists of Business (B), and Factory (Low Hazard) occupancy and use groups.

7-2.3 APPLICABLE CODES, STANDARDS, AND GUIDELINES

See Section 1-3.

7-3 FUNCTIONAL AND TECHNICAL REQUIREMENTS

7-3.1 Fire Sprinkler System

The building must be provided with complete wet pipe automatic sprinkler protection with full coverage throughout the facility in accordance with Section 9-7.2.1.1 UFC 3-600-01. Accordingly, all single-story Type I or II construction facilities greater than 15,000 feet² (1,394 m²) gross floor area must be provided with sprinkler system. The occupancy classification for sprinkler system in offices, administration, and public areas will be light hazard and for electrical and mechanical rooms must be considered as ordinary hazard occupancies. Facilities requiring sprinkler protection must be provided with systems that are designed using the area/density method of NFPA 13, except the discharge requirements for non-storage occupancies must be in accordance with Table 9-3, unless otherwise specified in the UFC 3-600-01.

7-3.2 Fire Flow

In accordance with Section 9-2.1, the fire flow provided for sprinklered buildings must be 1000 gpm at 20 psi. Per the latest test results, the fire flow for B3524 narrowly meets the code requirements. This issue must be further investigated and analyzed on the succeeding design and engineering stages.

7-3.3 Fire Department Vehicle Access

In accordance with UFC 3-600-01, Section 9-1, all Facilities greater than 5000 ft² (465 m²), or more than two stories in height, must have at least one means of all-weather ground access to allow emergency vehicles unimpeded access to the Facility. All weather ground access roads are present functioning properly.

7-3.4 Service Lateral Size

In accordance with Section 9-3.3.2, minimum size Service Lateral for fire sprinkler systems must be not less than 6-inches. (150 millimeters) in diameter.

7-3.5 Fire Department Connection

Facilities with fire department connections for fire suppression systems must be provided with suitable all-weather ground access surface for any apparatus within 150 feet (45 m) of such fire department connections. In addition, at least one hydrant must be located within 150 feet (45 m) of the fire department connection.

7-3.6 Connection to Fire Alarm System

The fire sprinkler system must be connected to the fire alarm system for transmission of alarms, supervisory and trouble signals. Fire sprinkler system water control valves and system riser must be provided with electric supervisory devices for monitoring the status

of devices and must be reported to the building fire alarm system for potential system tampering as well as water flows indicating fire alarm in the system.

7-3.7 Backflow Prevention

The backflow prevention device must be provided in accordance with UFC 3-600-01 Section 9-6.6. The horizontal backflow preventer must be installed so that the bottom of the assembly is no greater than 24 in. above the finished floor. For vertical backflow devices, upper operating handwheel must be no more than 6 feet above the finished floor. The hydraulic pressure loss due to presence of the backflow device must be accounted in the hydraulic calculations.

7-3.8 Hydraulic Calculations

The preparer of the shop drawings must perform hydraulic calculations in accordance with the applicable NFPA standard (NFPA 13), demonstrating that the design will provide an adequate supply for the fire suppression systems.

7-3.9 Backflow Device Test Heade

The new fire sprinkler systems must include test valves installed downstream of the backflow preventer for the purpose of backflow forward flow tests directed by NFPA standard. These valves must be angle or globe valves with 2.5-inch male National Standard Hose Threads with cap and chain. For each 250-gpm system flow rate, one 2.5-inch valve must be provided.

7-3.10 Seismic Design

The QFPE must coordinate with the Structural Engineer to determine the proper seismic design category for the project, in accordance with the IBC or ASCE guidelines. Seismic restraint is not required for Seismic design category A or B, except as otherwise required in UFC 3-600-01. For additional data, UFC 3-301-01 must be consulted.

7-3.11 Fire Sprinklers

Fire Sprinkler systems must be provided in the new electrical and mechanical rooms regardless of the fire resistance rated separation. The sprinkler k-factor must not be less than k-5.6 for the light hazard areas, including offices and admin rooms, and k-8.0 for all ordinary hazard areas, counting electrical and mechanical rooms, for the proposed Wet system.

7-3.12 Portable Fire Extinguisher

Must be provided in accordance with UFC 3-600-01, NFPA10, and NFPA101 requirements.

7-3.13 Fire Alarm and Mass Notification System

A complete and code compliant Mass Notification System must be provided in accordance with UFC 4-021-01. Accordingly, this standard applies to all DoD Components, and to all DoD inhabited buildings. All DoD components are required to provide individual building mass notification capability. Additionally, the UFC 4-021-01 applies to all new constructions with the applicability requirements of UFC 4-010-01.

7-3.14 HVAC Duct Smoke Detection System

The installation of a new HVAC system for the new B3524 will necessitate incorporation of duct smoke detector systems in compliance with the requirements of NFPA 90A Standard for the Installation of Air-Conditioning and Ventilating Systems. In accordance with Section 11.4.2.1 of NFPA 90A, smoke detectors listed for use in air distribution systems must be located downstream of the air filters and ahead of any branch connections in all air supply systems having a capacity greater than 2,000 CFM and must automatically de-energize their respective fans upon detecting the presence of smoke.



7-3.16 Building Egress System

The new Building 3524 egress system must comply with the requirements of NFPA 101 and UFC 3-600-01 requirements.

7-3.17 Applicability of Building and Life Safety Codes

The occupancy classification as it relates to fire/smoke resistance rating of interior non-load bearing partitions (other than occupancy separation), means of egress, interior finish, features of fire protection (including vertical openings) and associated requirements must be compliant with requirements of NFPA 101. The IBC must be utilized to determine the occupancy classification as it relates to allowable construction type, building height, building area, building separation distance, occupancy separation and associated requirements.

7-3.18 Cybersecurity

In accordance with Section 1-13 UFC 3-600-01 all control systems (including systems separate from an energy management control system) must be planned, designed, acquired, executed, and maintained in accordance with DoD Instruction 8500.01, DoD Instruction 8510.01, and as required by individual Service Implementation Policy.

7-3.19 Hazardous Materials

Requirements of Section 3-4 of UFC 3-600-01 must apply. For the purpose of occupancy classification, the maximum allowable quantities (exempt amounts) noted in NFPA 400 must be used except for ammunition and explosives, for which the maximum allowable quantity noted in IBC Chapter titled "Occupancy Classification and Use" must be used.

7-3.20 Surge Suppression (SPD)

Provide primary power in accordance with Provide SPD on all 120 Vac circuits to control panels, subpanels, transmitters, amplifier panels, and booster panels. SPD must have both a UL 1449 and UL 1283 listing and must be located in an adjacent hinged terminal box.

7-3.21 Power

Provide primary power in accordance with UFC 3-520-01. This includes the provision of a lock-on circuit breaker. For secondary power, provide rechargeable batteries per NFPA 72 to operate the fire alarm system under supervisory conditions for 48 hours and all alarm devices for an additional 15 minutes. Where the fire alarm system also serves as a mass notification system, refer to UFC 4-021-01 for additional requirements.

7-4 DESIGN OBJECTIVES AND PROVISIONS

A complete fire sprinkler system in accordance with requirements of NFPA13, IBC, and UFC 3-600-01 must be provided throughout the building. The earthquake sway bracing for riser and overhead piping must be provided.

A complete fire alarm and MNS system in accordance with requirements of NFPA70, 72, IBC, UFC 3-600-01, UFC 4-021-02, and UFC must be provided.

A duct smoke detection system for new HVAC duct smoke monitoring per requirements of NFPA 90A and UFC 3-600-01 must be provided.

Interconnection between fire sprinkler system alarm and supervisory equipment with building fire alarm system in compliance with NFPA and UFC requirements must be provided.

The preparer of the shop drawings must perform hydraulic calculations in accordance with the NFPA13 standard to demonstrate that available water supply source can provide an adequate amount of flow and pressure to cover fire suppression system demand needs.

In addition, all items indicated in previous Section 7-3 must also be considered in design objectives.

7-5 COORDINATION WITH INSTALLATION OR OUTSIDE AGENCIES

The fire protection scope of work requires coordination between the fire sprinkler system supervision and alarm equipment with building fire alarm and monitoring system, and with air base remote supervising and fire stations, for transmitting alarm, supervisory, and trouble signals. The sprinkler piping, sway bracing, and hangers require coordination with structural, electrical raceways, mechanical AHU, and ductwork construction.

APPENDIX A – CONCEPT DRAWINGS

COLOR LEGEND

-  RESTROOMS
 STORAGE
 BREAK AREA
 MECHANICAL / ELECTRICAL RM.

[illegible]

DESIGNED BY: AIR FORCE INSTALLATION AND MISSION SUPPORT CENTER DETACHMENT 2 (PAOAF) JBPHH, HI

DRAWN BY: MA

QUOTATION NO.: FA21523F0011

CONTRACT NO.: FA589032100012

CREATED BY: EG

SUBMITTED BY: 3500 CDM Smith JV LP 3500 Parkway Lane Suite 500 Peachtree Corners GA 30092

APPROVED BY: [Signature] DATE: 08 SEP 2024

PROJECT NO.: AFD #1230614

KADENA AIR BASE
OKINAWA, JAPAN
REPAIR SOW OPS, B3524
PACAF FSRM (DNA) TRANCHE 1

SHEET ID
B3524
SD101

35% SUBMITTAL

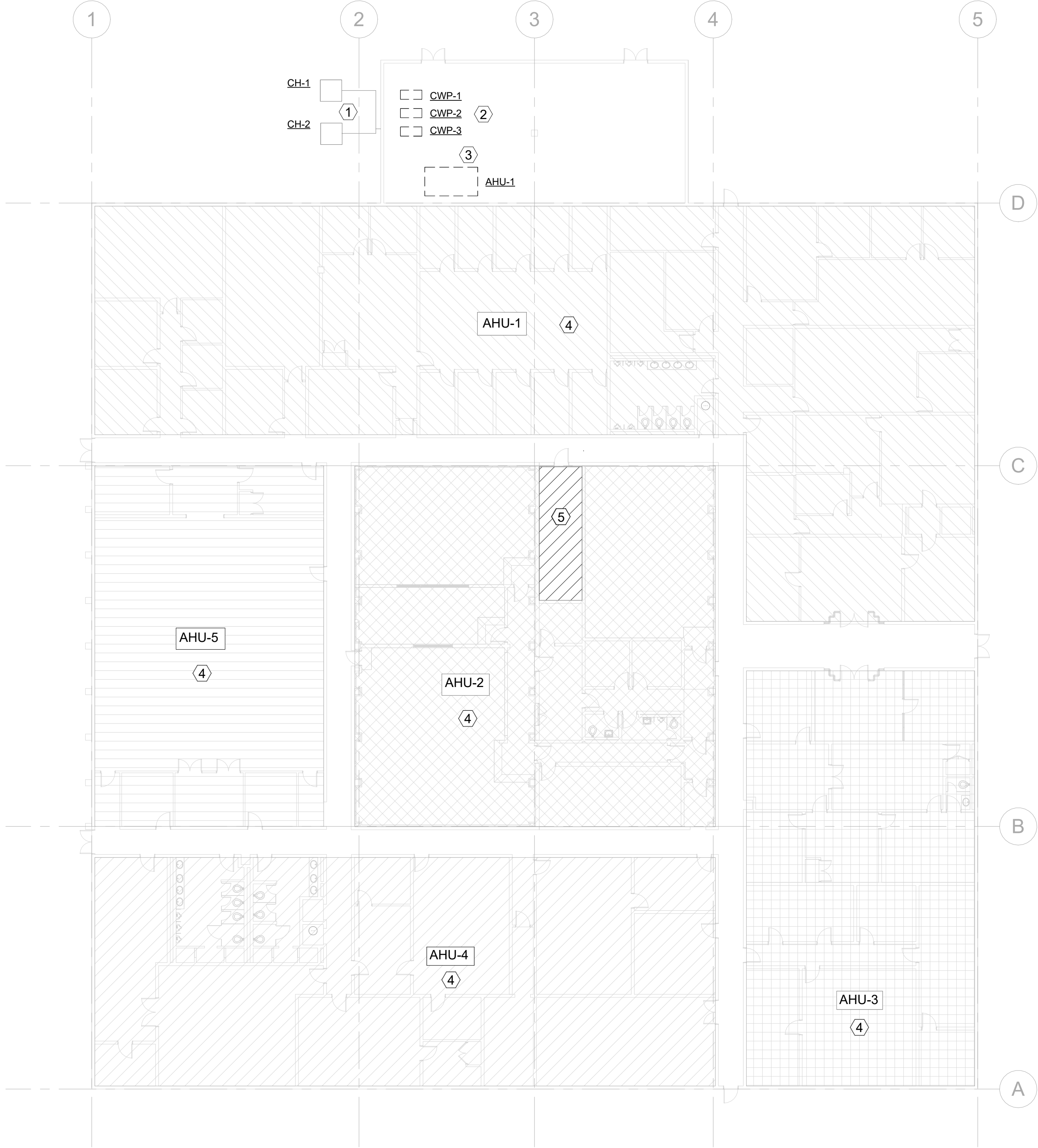
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EXISTING EQUIPMENT SCHEDULE-OLD				
MARK	MANUFACTURER	MODEL	SIZE	AGE
AHU-1	CARRIER	39MN21	10,300 CFM	2005
AHU-2	CARRIER	39LA21	11,000 CFM	2011
AHU-3	CARRIER	39MN10	5,000 CFM	2005
AHU-4	CARRIER	39MN10	5,000 CFM	2005
AHU-5	CARRIER	39MN10	5,000 CFM	2005
CH-1	CARRIER	30RAP060	60 TONS	2023
CH-2	CARRIER	30RAP060	60 TONS	2023
CWP-1	TACO	F11509	110@65', 5HP	2019
CWP-2	TACO	F11509	110@65', 5HP	2019
CWP-3	TACO	F11509	110@65', 5HP	2005



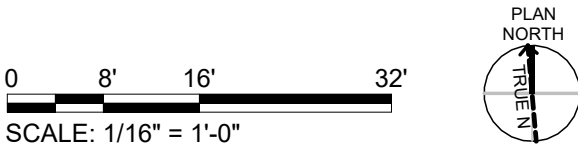
A4 HVAC FLOOR PLAN DEMOLITION
SCALE: 1/16" = 1'-0"

SHEET NOTES

1. EACH HVAC UNIT SERVES THE HATCHED AREA INDICATED.

KEYNOTES

1. THE EXISTING CHILLERS (CH) SHALL REMAIN AND BE REUSED. REMOVE ALL EXTERIOR CHILLED WATER PIPING.
2. REMOVE EXISTING CHILLED WATER PUMPS (CWP) AND ALL CHILLED WATER PIPING.
3. REMOVE EXISTING AIR HANDLING UNIT (AHU) AND ALL ASSOCIATED DUCTWORK.
4. THE ROOF STRUCTURE IS BEING REMOVED. ALL INTERIOR CHILLED WATER PIPING, DUCTWORK, CONTROLS AND ALL APPURTENANCES ASSOCIATED WITH THE HVAC SYSTEM SHALL BE REMOVED.
5. EXISTING INFORMATION TRANSFER NODE (ITN) ROOM TO REMAIN DURING COA-1 AND COA-2 WORK. PROVIDE TEMPORARY HVAC DURING BUILDING RENOVATION WORK.

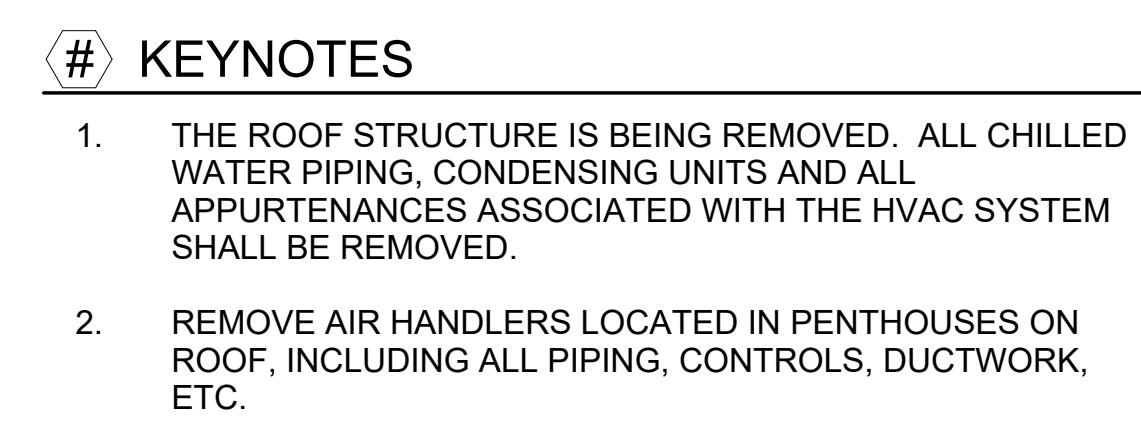


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CHECKED BY: ALR	CONTRACT NO.: FA890321D0012
SUBMITTED BY: EMG	SIZE: ANSI D
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POND CDM Smith JV LP 9500 Parkway Lane Suite 500 Atlanta, Georgia 30092 POND #1230613	

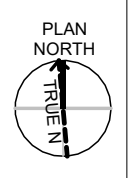
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OKINAWA, JAPAN
REPAIR SQ OPS FACILITY: B3524
PACAF PSRW (DNA) TRANCHE 1
HVAC FLOOR PLAN DEMOLITION

SHEET ID
B3524
MD101



0 8' 16' 32'

SCALE: 1/16" = 1'-0"



KADENA AIR BASE
OKINAWA, JAPAN
REPAIR SQ OPS FACILITY, B3524
PACAF FSRM (DNA) TRANCHE 1

HVAC ROOF PLAN DEMOLITION

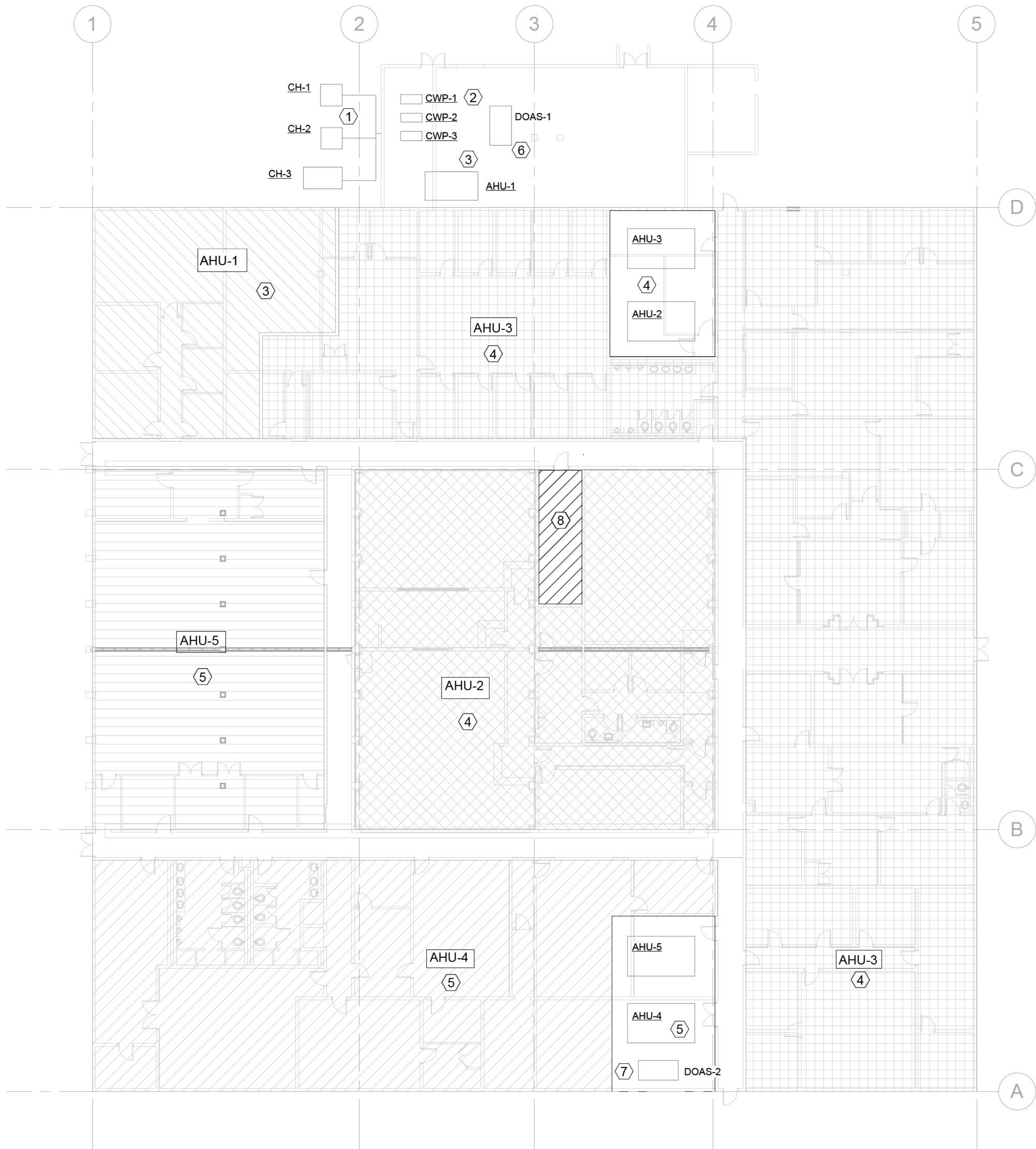
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B3524
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JBPBHQ	DRAWN BY:	SOLICITATION NO.:
	CYAN	FAG21527-P0011
	COMMITTED BY:	FA89032-D0012
	AIR	
POND CDM SMITH JV LP	RECEIVED BY:	
5500 Parkway Lane	EMG	
Suite 500	ASND	
CONCORD CORNERS		
Georgia 30092		

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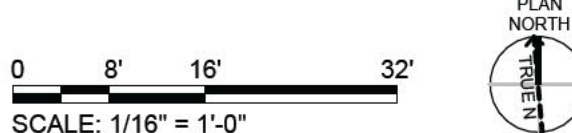
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A4 HVAC FLOOR PLAN NEW WORK
SCALE: 1/16" = 1'-0"

- # KEYNOTES**
- EXISTING CHILLERS, INSTALLED IN 2023, ARE TO BE REUSED. PROVIDE NEW CHILLER(S) AS NECESSARY SO THAT THE BUILDING WILL HAVE FULL REDUNDANCY IN CHILLED WATER CAPACITY.
 - NEW CHILLED WATER PUMPS. ROUTE NEW CHILLED WATER PIPING FROM PUMPS TO ALL NEW AIR HANDLERS.
 -
 - NEW AHU-2 & 3 IN NEW MECHANICAL ROOM, TO SERVE THE JOAC AND THE HQ AREAS.
 -
 - NEW DEDICATED OUTSIDE AIR SYSTEM, DOAS-1. THIS UNIT WILL SUPPLY PRE-CONDITIONED OUTSIDE AIR TO AHU-1, 2 & 3.
 - NEW DOAS-2. THIS UNIT WILL SUPPLY PRE-CONDITIONED OUTSIDE AIR TO AHU-4 & 5.
 - ALL NEW OR EXISTING COMM ROOMS SHALL HAVE FULL REDUNDANCY AND BE SERVED WITH BOTH DUCTLESS SPLITS AND A VAV BOX FROM THE AREA'S AIR HANDLER.



MARK		DESCRIPTION	DATE

DESIGNED BY:	ISSUE DATE:
MPM	06 SEP 2024

DRAWN BY:	SOLICITATION NO.:
GVL	FA521523F0011

CHECKED BY:	CONTRACT NO.:
ALR	FA890321D0012

SUBMITTED BY:	SIZE:
EMG	A3/D

AIR FORCE INSTALLATION AND MISSION SUPPORT CENTER DETACHMENT 2 (PACAF) JBPHH, HI

POND CDM SMITH JV LP 3500 Parkway Lane Suite 300 Jacksonville, Florida 32202 Georgia 30092

POND CDM Smith JOINT VENTURE POND #230613

KADENA AIR BASE
OKINAWA, JAPAN
REPAIR SQ OPS FACILITY: B3524
PACAF FSRW (DNA) TRANCHE 1

HVAC FLOOR PLAN NEW WORK

SHEET ID
B3524
M-101



SHEET ID
B3524
PD101

FINAL 35% DESIGN

A4 PLUMBING FLOOR PLAN NEW WORK
SCALE: 1/16" = 1'-0"

SCALE: 1/16" = 1'-0"

KEYNOTES

1. PROPOSED NEW MEN'S AND WOMENS RESTROOM LOCATION. PROVIDE NEW WATER HEATER IN A CLOSET IN THE SAME AREA AS NEW RESTROOMS.
2. PROVIDE NEW DOMESTIC WATER ENTRANCE IN SAME LOCATION AS EXISTING. EXTEND TO NEW SET OF RESTROOMS AND PROVIDE MAKE-UP WATER FOR THE CHILLED WATER SYSTEM.
3. PROVIDE NEW SANITARY SEWER ENTRANCE IN SAME LOCATION AS EXISTING. SANITARY LINE SHALL SERVE THE MECHANICAL ROOM.
4. PROVIDE NEW SANITARY SEWER ENTRANCE IN SAME LOCATION AS EXISTING. SANITARY LINE SHALL SERVE THE NEW SET OF RESTROOMS.
5. PROVIDE NEW SANITARY SEWER ENTRANCE. SANITARY LINE SHALL SERVE THE RESTROOM.
6. NEW HVAC EQUIPMENT IN EXISTING OR NEW MECHANICAL ROOM. PROVIDE FLOOR DRAINS.

[illegible]

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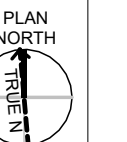
POND CDM Smith
JOINT VENTURE
POND #1230613

POND CDM SMITH
3500 Parkway Lane
Suite 500
Peachtree Corners
Georgia 30092

MISSION SUPPORT CENTER
DETACHMENT 2 (PACAF)
JBPHH, HI

OKINAWA, JAPAN
REPAIR SQ OPS FACILITY, B3524
PACAF FSRM (DNA) TRANCHE 1

SHEET ID
B3524
P-101



FINAL 35% DESIGN

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DRAWN BY: JSH		
CHECKED BY: JSH		
CONTRACT NO.: FA580321D0012		
SOLICITATION NO.: FA521523F0011		
RMR		
ISSUED BY: JSH		
ISSUED DATE: 06 SEP 2024		
AIR FORCE INSTALLATION AND MISSION SUPPORT CENTER (DETACHMENT) (AFACF) JBPB.HI		

KADENA AIR BASE
OKINAWA ISLAND, JAPAN
BUILDING B3524
PACAF FSRM EAS TRANCHE 1

SHEET ID
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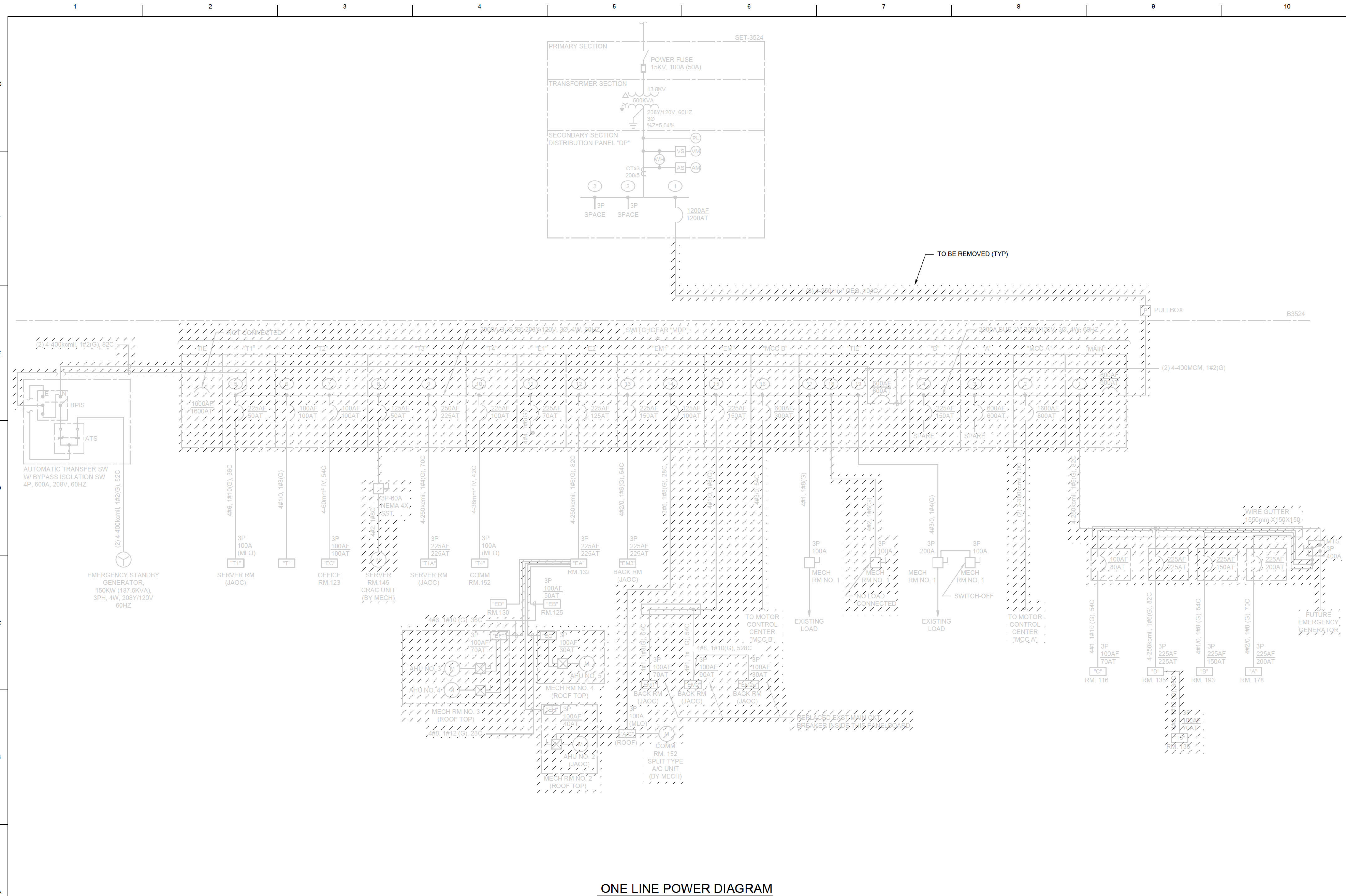


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	CHECKED BY:	CSH	CONTROL NO.:	FA3889031D0012
	SUBMITTED BY:	BAE		
3500 Parkway Lane Suite 500 Peachtree Corners Georgia 30092		NYS/D POND # 230558		

SHEET ID
B3524
E-101





ONE LINE POWER DIAGRAM

[illegible]

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	CHECKED BY:	USH	CONTRACT NO.:	FA589032 D0012
	SUBMITTED BY:	USH		
POND 2 15400 Parkway Lane Suite 500 Peachtree Corners Georgia 30092 JOINT VENTURE POND#2320558			DATE: BY: ANSI D	

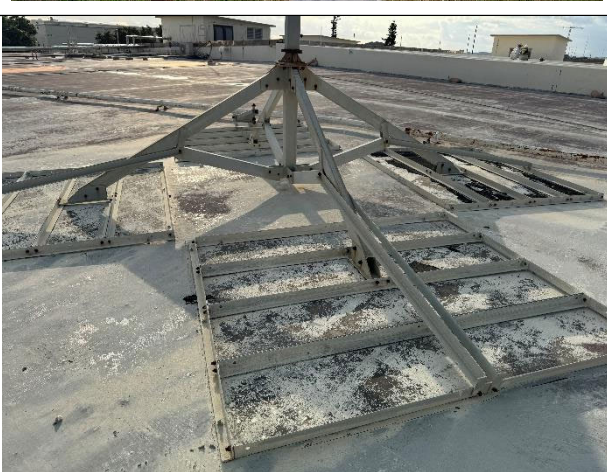
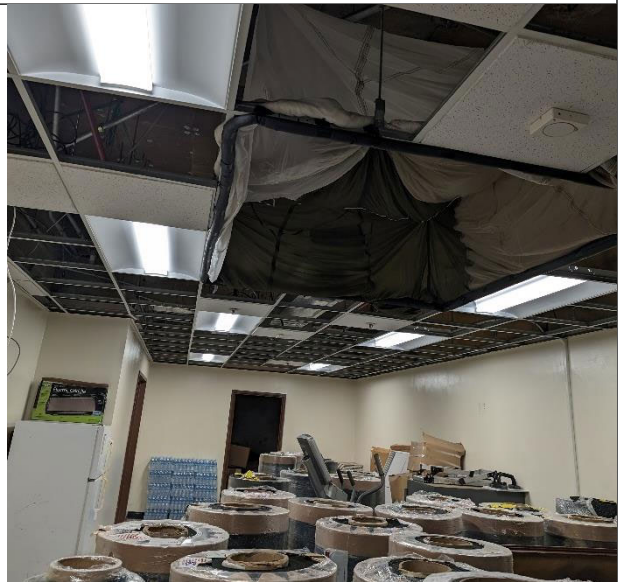
KADENA AIR BASE
OKINAWA ISLAND, JAPAN
BUILDING B3524
PACAF FSRM EAS TRANCHE 1

SHEET ID

ED-601

APPENDIX D – SITE VISIT PHOTOGRAPHS

Appendix D – Site Visit - Photographic Documentation



APPENDIX E – HYDRANT FLOW TEST



Hydrant Flow Test Report

Project Name: PACAF - KADENA TRANCHE 1 -SOG OPS FACILITY, B3524 Project Number: 282576

Location: See the attached maps Date: 01/24/2024

Test made by: Patrick Veldenian, PE, and Dan Brace, PE Time: 14:00

Representative of: CDM Smith and POND

Witness: Kadena AFB Water Shop Personnel Contact info: _____

Flow hydrants: (Hyd. ID #1) 21-50 (Hyd. ID #2) _____ (Hyd. ID #3) _____ Hyd. ID #4) _____

Static / Residual hydrant (Hydrant ID): 21-58

Hydrant Locations (Street address): Hydrant #21-50 N-W and #21-58 W of the B3524

Static (psi): 92 psi Residual (psi): 44 psi

Pitot reading (psi): 26 psi Water flow (GPM): 856 $Q = 29.84cd^2 \sqrt{p}$ [NFPA 291; 4.7.3]

Size nozzle (in.): 2.5" Discharge coefficient (circle one): [0.9] [0.8] [0.7]

Elevation difference from finished floor to test hydrant (ft.): Approximate 1 ft.

Size of underground mains (in.): 6 or 8 inches

Water distribution (circle one): [Dead end] [Looped]

Location and type of water source (Elevated water tank etc.) (Include any additional supplemental water supply sources): The base water distribution system is served by wells, elevated tanks, and pressure booster pump station.

AHJ contacted / informed of flow test (circle one): [yes] [no] Contact info: edwin.salazar.1@us.af.mil

PPE (circle one): [yes] [no]

Hydrant closed fully (circle one): [yes] [no]

Hydrant caps have been re-attached (circle one): [yes] [no]

Did hydrant drain after closure (circle one): [yes] [no]

Was there sufficient drop in pressure as required by NFPA 291; 4.3.6 (Min 25% or total demand flow): [yes] [no]

Note: Provide location map where available. Label map with test and hydrant location and ID's.

Remarks:

Significant amount of scale, stain, and rust was observed during the hydrant flushing.

Flow at 20 psi: 1,066 gpm

